

Electronic Design

Name:

Oumar DIAKITE

November 2025

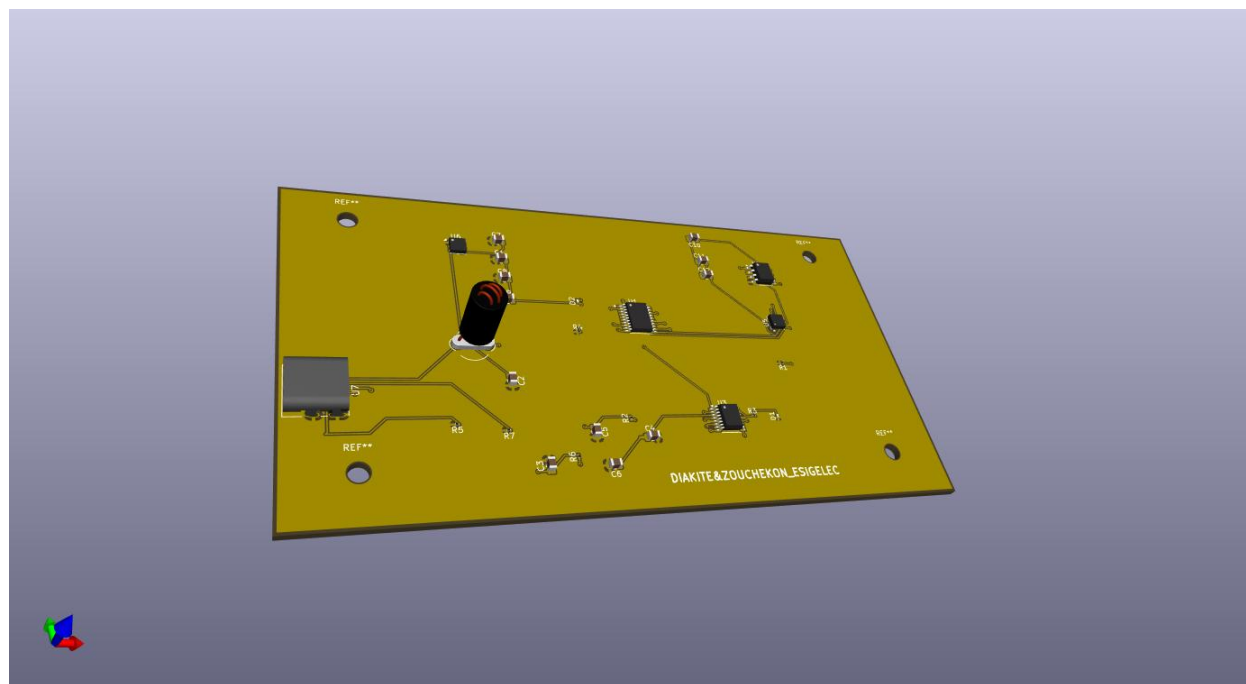
1. Project Description

Our circuit consists of an STM32F070F6 microcontroller, a USB-C port, an inductor, resistors, capacitors, indicator LEDs, a voltage regulator, an ADXL363BCCZ-RLZ accelerometer, an AT25SF081-SSHD memory chip, and an FT230XS USB-to-UART converter. In this circuit, our accelerometer represents the "sensor" block. It is a 3-axis digital accelerometer that also includes a temperature sensor and an ADC for an external signal. It uses SPI (Serial Port Interface) communication and provides a 12-bit resolution for each axis. This sensor also features an adjustable sampling rate ranging from 12.5Hz to 400Hz. Furthermore, it has an internal FIFO (First-in/First-out) of 512 samples, which is useful for collecting motion data without continuously loading the microcontroller.

Our microcontroller is a 32-bit ARM Cortex. It has a clock speed of 48 MHz and 4 KB of SRAM (static random-access memory). Several communication protocols are supported by this microcontroller: SPI, I2C, and USART. 16-bit timers can also be used. Like most microcontrollers, this component also includes a 12-bit ADC. Such a microcontroller requires non-volatile memory within its circuit architecture.

The AT25SF081 is a Flash memory with a capacity of 8Mbit and a maximum SPI bus frequency of 104MHz. Like all non-volatile memory, it can be used to store firmware, logs, configurations, etc.

2. Final Design



3. Difficulties & Challenges

My difficulties are mostly related to the prerequisites in electronics.

It was my first time encountering certain electronic concepts related to different pins, such as CTS and RTS. I didn't know how to connect such pins. Often, some datasheets offer guidance, while others don't. Also, adding capacitors on both sides to limit current and prevent sudden temperature drops seems like an important prerequisite.

In summary, a basic understanding of electronics, particularly regarding safety and the various communication protocols within a circuit, can be invaluable for good PCB design. When knowledge is limited, there's a high risk of damaging many components. Another challenge was the use of KiCad.

4. Conclusions

The sessions were good overall. The professor was very attentive to the students.

I learned new things about circuit safety (putting capacitors in parallel, using a regulator, etc.). Furthermore, I was able to edit a symbol and create a new library associated with it. Regarding PCB design, I strengthened my routing skills. I better understood the purpose of certain layers. As for improvements, it might be worthwhile to create a beginner's guide for new software, whether it's KiCad or another CAD program. Similarly, expanding on the specifications document wouldn't be a bad idea. For example, we could define the dimensions of the grids, traces, and so on. This would certainly add complexity. However, it would be an opportunity to use and encounter issues with .NET classes (default sizes, properties, etc.).